

The Dot Product

ANSWER KEY



Computing dot products and angles

1 Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:

a $\mathbf{u} \cdot \mathbf{v} = (2)(-3) + (5)(4) = 14$.

b $|\mathbf{u}|$ and $|\mathbf{v}|$ $|\mathbf{u}| = \sqrt{29}$, $|\mathbf{v}| = 5$.

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place $\cos \theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7^\circ$.

2 Determine whether each pair of vectors is orthogonal:

a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle$ $12 - 12 = 0$: orthogonal. **b** $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$ $2 + 2 = 4 \neq 0$: not orthogonal.

3 Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.

$2k - 12 = 0$, so $k = 6$.

Projections and work

4 Find the vector projection of $\mathbf{u} = \langle 6, 2 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v} = \frac{18}{9} \langle 3, 0 \rangle = \langle 6, 0 \rangle$.

5 A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.

$W = \mathbf{F} \cdot \mathbf{d} = 80$ J.

6 A wagon is pulled a distance of 20 m by a rope making an angle of 30° with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.

$W = |\mathbf{F}||\mathbf{d}|\cos \theta = 50(20)\cos 30^\circ \approx 1000(0.8660) \approx 866.0$ J.

Properties of the dot product

7 Let $\mathbf{u} = \langle 3, 1 \rangle$, $\mathbf{v} = \langle -2, 4 \rangle$, $\mathbf{w} = \langle 1, 1 \rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).

$\mathbf{v} + \mathbf{w} = \langle -1, 5 \rangle$: $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = -3 + 5 = 2$. Separately: $\mathbf{u} \cdot \mathbf{v} = -6 + 4 = -2$, $\mathbf{u} \cdot \mathbf{w} = 3 + 1 = 4$; sum = 2. Both equal 2. ✓

8 Show that $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$ for $\mathbf{u} = \langle 4, -3 \rangle$.

$\mathbf{u} \cdot \mathbf{u} = 16 + 9 = 25$. $|\mathbf{u}|^2 = (\sqrt{25})^2 = 25$. Equal, as expected.

Angle classification

- 9 Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:
- a $\langle 3, 5 \rangle$ and $\langle 2, -1 \rangle$ Dot product $= 6 - 5 = 1 > 0$: acute.
- b $\langle -1, 4 \rangle$ and $\langle 8, 2 \rangle$ Dot product $= -8 + 8 = 0$: right angle.
- c $\langle 2, 2 \rangle$ and $\langle -3, -3 \rangle$ Dot product $= -6 - 6 = -12 < 0$: obtuse.
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Scalar projection and further applications

- 10 Find the scalar projection of $\mathbf{u} = \langle 5, -2 \rangle$ onto $\mathbf{v} = \langle 3, 4 \rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$), to two decimal places.
 $\mathbf{u} \cdot \mathbf{v} = 15 - 8 = 7$. $|\mathbf{v}| = 5$. $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{7}{5} = 1.40$.
- 11 A sled is pulled by a rope at 40° above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.
 $W = |\mathbf{F}|\cos(40^\circ) \cdot d = 120\cos 40^\circ \cdot 15 \approx 120(0.766)(15) \approx 1378.8 \text{ J}$.
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Decomposing a vector into parallel and perpendicular parts

- 12 Decompose $\mathbf{u} = \langle 7, 4 \rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0 \rangle$ and a component perpendicular to $\mathbf{v}$.
 Parallel: $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{7}{1}\langle 1, 0 \rangle = \langle 7, 0 \rangle$. Perpendicular: $\mathbf{u} - \langle 7, 0 \rangle = \langle 0, 4 \rangle$. Check: $\langle 7, 0 \rangle + \langle 0, 4 \rangle = \langle 7, 4 \rangle$ ✓, and $\langle 0, 4 \rangle \cdot \langle 1, 0 \rangle = 0$ ✓.
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Deriving the dot product angle formula

- 13 State the formula relating the dot product to the angle between two vectors, and explain why $\cos \theta = 0$ corresponds to orthogonal vectors.
 $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \theta$. Since $|\mathbf{u}|, |\mathbf{v}| > 0$ for nonzero vectors, the dot product is zero exactly when $\cos \theta = 0$, i.e. $\theta = 90^\circ$ — perpendicular vectors.
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The Cauchy-Schwarz inequality (preview)

- 14 Explain why $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \theta$.
 Since $-1 \leq \cos \theta \leq 1$ for any angle, multiplying by the non-negative quantity $|\mathbf{u}||\mathbf{v}|$ gives $-|\mathbf{u}||\mathbf{v}| \leq \mathbf{u} \cdot \mathbf{v} \leq |\mathbf{u}||\mathbf{v}|$, which is exactly $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$.
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Dot products with variable components

- 15** Find all values of k such that $\langle k, 3 \rangle$ and $\langle k, -3 \rangle$ are orthogonal.
Dot product: $k^2 - 9 = 0 \Rightarrow k = \pm 3$.
- 16** Find all values of k such that the angle between $\langle 1, k \rangle$ and $\langle 2, 1 \rangle$ is 45° .
 $\cos 45^\circ = \frac{2+k}{\sqrt{1+k^2} \cdot \sqrt{2}} = \frac{\sqrt{2}}{2}$. Squaring: $\frac{(2+k)^2}{5(1+k^2)} = \frac{1}{2}$, so $2(2+k)^2 = 5(1+k^2)$: $2(4+4k+k^2) = 5+5k^2$,
 $8+8k+2k^2 = 5+5k^2$, $3k^2-8k-3=0$. Quadratic formula: $k = \frac{8 \pm \sqrt{64+36}}{6} = \frac{8 \pm 10}{6}$: $k = 3$ or $k = -\frac{1}{3}$.

A capstone dot product application

- 17** A weather vane experiences wind force $\mathbf{F} = \langle 15, -8 \rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4 \rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F} \times \mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.
Scalar cross product: $F_x d_y - F_y d_x = 15(4) - (-8)(3) = 60 + 24 = 84$. Perpendicular component magnitude $= \frac{84}{5} = 16.8$ N. Parallel component via dot product: $\mathbf{F} \cdot \mathbf{d} = 45 - 32 = 13$; magnitude $= \frac{13}{5} = 2.6$ N. Together these decompose the total force into turning and non-turning parts.
- 18** Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10 \rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45 \rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).
 $\mathbf{s}_1 \cdot \mathbf{s}_2 = 40(15) + 25(30) + 10(45) = 600 + 750 + 450 = 1800$. This dot product is a single combined measure of alignment/overlap in their sales patterns, but it does NOT directly reveal who sold more in total (that would come from summing each vector's own components: 75 vs 90 units) — the dot product measures a different kind of relationship (similarity in distribution across categories), not total volume.